

RedMMOTitan — Developer Handoff (EN)

2026-08-07. Collaborator sanyarud@gmail.com → GitHub sanyarud (write, pending). Netcode/PvP: NETWORKING_PVP.md. PPG/PlanetGen optional via TitanFundamentals.

START HERE — RedMMOTitan developer handoff

Generated: 2026-08-07

Ukrainian: START_HERE_UK.md · PDF UK: RedMMOTitan_Developer_Handoff_UK.pdf

PDF (EN): RedMMOTitan_Developer_Handoff.pdf

Full handoff: DEVELOPER_HANDOFF.md

Networking / PvP (primary for this developer): NETWORKING_PVP.md · NETWORKING_PVP_UK.md

Fab / Marketplace checklist: FAB_MARKETPLACE_INVENTORY.md

PPG / PlanetGen-free start: PPG_PLANETGEN_FREE_START.md

Collaborator: sanyarud@gmail.com → GitHub sanyarud (write/push invites pending accept).

60-second path

- 1 Clone the standalone handoff repo (see clone URL in GITHUB_ACCESS.md).
- 2 Install Unreal Engine 5.8 (exact association in Titan.uproject / TitanFundamentals.uproject).
- 3 Do not wait on PPG or PlanetGen for first compile / first PIE:
 - Open TitanFundamentals.uproject
 - Default map: /Game/ThirdPerson/Maps/ThirdPersonMap
 - Build TitanEditor Win64 Development, then PIE
- 1 When marketplace planet plugins are available, switch to full Titan.uproject or the clean D-drive RedMMO.uproject for planetary work.
- 2 Read DEVELOPER_HANDOFF.md before changing locomotion, fused terrain, or production maps.

Clone → open → build → PIE

```
git clone <CLONE_URL_FROM_GITHUB_ACCESS.md>
cd RedMMOTitan-DeveloperHandoff
```

```
# Fundamentals (no PPG / PlanetGen required)
start TitanFundamentals.uproject
```

In Visual Studio / Rider / Unreal Build:

```
Target: TitanEditor
Platform: Win64
Configuration: Development
```

PIE checklist (fundamentals baseline):

Check	Expected
Editor opens ThirdPersonMap	No missing PPG/PlanetGen hard fail
PIE possess	Character moves with WASD + mouse
Compile RedMMO module	Succeeds with FocalRig/WorldGen disabled
No second editor	Only one UnrealEditor at a time on this machine

PIE checklist (full planetary / clean RedMMO — optional plugins required):

Check	Expected
Wheel	Thrust (recent RedMMO editor fix)
Atmosphere exit	Works (recent fix)
Weapons / plumes / jetpack	Present on live RedMMO editor build
Terrain blend	Present
Running animation	Do not overwrite / "fix" without explicit authority

Machine paths (owner workstation)

Role	Path
Repo / knowledge root	D:\RedMMOTitan
Clean planetary project	D:\RedMMOTitanWindowsData\Projects\RedMMO\RedMMO.uproject
Accepted packaged R92 baseline	D:\RedMMOTitanWindowsData\Builds\RedMMO_R92_Playable_20260806T1740Z\Windows\RedMMO.exe
This diagnostics pack	D:\RedMMOTitanWindowsData\Diagnostics\RedMMO_DeveloperHandoff_20260807\
R92 agent takeover pack	D:\RedMMOTitanWindowsData\Diagnostics\RedMMO_R92_AgentTakeover_20260806T1805Z\
Dependency disposition	D:\RedMMOTitanWindowsData\Diagnostics\R92_DependencyUnblock_20260807_000731\R92_DEPENDENCY_DISPOSITION.md
Engine	D:\UE_5.8

Do not use C:\Users\user\Documents\Red MMO Windows.

Protected items (do not casually touch)

- Running / locomotion anim graph on ABP_RedTrooperFemale — treat as protected; recent PIE work must not regress run.
- Protected fused 27/6 authoring inputs and hashes (see ProjectKnowledge / handoff).
- Production map hashes and R92 package tree identity.
- Never `git reset --hard` the dirty owner worktree on D:\RedMMOTitan.

Next fundamentals work (priority)

- 1 Stabilize on-foot locomotion feel without breaking the protected run anim.
- 2 Weapon / jetpack / ship entry-exit feel on a map that opens without PPG if planetary plugins are missing.
- 3 Then (with PPG or PlanetGen restored): home-world art, shore bands, multiplayer, fused-consumer binding.

Continue with DEVELOPER_HANDOFF.md.

Networking / Replication / Multiplayer / PvP — programmer start

Audience: gameplay / netcode programmer (listen-server, replication, PvP).

Ukrainian: NETWORKING_PVP_UK.md

PPG/PlanetGen: not required for first netcode PIE — use `TitanFundamentals.uproject` + `ThirdPersonMap`.

Current architecture (honest)

Layer	What exists	Proven?
Topology	Unreal listen-server (host is also a player). Dedicated server not accepted.	PIE 2-client listen-server used historically for aim/jetpack/fire replication
Transport	Steam Integration Kit (SIK) v1.9 as sole Steam stack (replacement <code>OnlineSubsystemSteam</code> + <code>SteamSockets</code>). Epic stock OSS Steam plugins disabled. Dev App ID 480 (Spacewar) via <code>steam_appid.txt</code>	Packaged smokes: Steam auth/relay + single-account host lifecycle. Real two-account Steam join/PvP not proven
Session UI	Escape → Multiplayer/Lobby; F8 runtime / F6 in PIE (editor owns F8). Create / Find / Join / Reconnect / Invite / Leave. Sessions advertise ≥8 players. No silent auto-host after empty search	UI + host lifecycle smoke; friend join open
Authority	Server-authoritative weapons heat/overheat, bolt spawn, ship pilot/weapons, damage; aim direction + jetpack flags replicated	Official-MCP two-client listen-server PIE verified aim/jetpack/fire/heat (NULL OSS path — gameplay replication, not Steam transport)
GameState	Celestial / PPG frame producers and registry work exist; real 1S/2C late-join transport + radar consumers open (M05)	Single-process ordering evidence ≠ multiplayer acceptance
PvP	<code>bIsEnemy</code> , health/shield/armor, downed, grapple-on-player pull, lethal ragdoll / ship death paths in character/ship code	Not a finished PvP mode; no ranked/matchmaking

Key classes (start here)

System	Primary files	Net notes
Player / jetpack / aim / weapons	<code>Source/RedMM0/RedPlayerCharacter.h/.cpp</code>	Large <code>DOREPLIFETIME</code> block: Health/Shield/Armor/Fuel, aim, weapon heat, jetpack flags, abilities, grapple/slam
Movement	<code>RedCharacterMovement.*</code>	Radial gravity / planet movement — test carefully under listen-server
Projectiles	<code>RedBolt.*</code> , <code>WeaponFirer.*</code>	Server-spawned bolts; presentation vs authority
Ship (StarSparrow)	<code>RedShip.*</code> , <code>RedShipMovementComponent.*</code>	Replicated Health, WeaponHeat, Pilot, landing flags
Shuttle	<code>RedShuttleBase.*</code>	Related craft occupancy / audio GC fix history
Sessions / Steam	<code>RedGameInstance.*</code> , <code>RedSessionBrowserWidget.*</code> , <code>RedHUD.*</code> (lobby open)	SIK session browser; App 480
Pause / lobby entry	<code>RedPauseMenuWidget.*</code>	Escape menu Multiplayer row
GameMode	<code>RedGameMode.*</code> , clean project <code>RedPPGGameplayGameMode.cpp</code>	Default modes differ Titan vs clean RedMMO
Celestial / MP consumers	<code>RedCelestialFrameRegistry.*</code> , GameState helpers	M05 still open for real 1S/2C
Focused MP tests	<code>Source/RedMM0/Tests/RedDEF0003TwoClientPIETests.cpp</code>	Listen-server count helpers — extend, don't treat as full Steam proof

Anim / aim presentation (protect run): Content/RedMMO/Characters/ABP_RedTrooperFemale — do not casually rewrite run/locomotion. Aim/jetpack pose work should stay additive / parallel. FocalRig marketplace absent → ControlRig substitute in RedMMOEditorTools.

Where to start (recommended order)

- 1 Day 1 — no Fab planets: Open TitanFundamentals.uproject, build TitanEditor, PIE ThirdPersonMap. Confirm compile with PlanetGen/PPG/FocalRig/WorldGen off.
- 2 Day 1–2 — listen-server gameplay replication: Editor → Play → Number of Players = 2, Net Mode = Play As Listen Server. Validate move/aim/fire/jetpack replication between windows (NULL OSS is fine for this gate).
- 3 Day 2–3 — PvP damage loop: Trace Health/Shield damage, bIsEnemy, downed, bolt hit authority in RedPlayerCharacter / RedBolt. Add a minimal deathmatch spawn/score only after replication is stable.
- 4 When ready for Steam: Enable/copy Plugins/SteamIntegrationKit, keep Epic OSS Steam disabled, Steam client logged in, App 480. Host on machine A, join on machine B with different Steam accounts. Do not claim success from single-account smoke.
- 5 Defer: PPG home world, PlanetGen maps, fused terrain, final art — not blockers for netcode fundamentals.

Known gaps (net / MP)

- Real 1 server + 2 clients Steam transport (initial/update/stale/late-join) unaccepted
- Dedicated server, persistence, matchmaking, Steam App ID beyond 480
- Radar/minimap multiplayer consumers
- Warp / interplanetary replication
- Packaged two-account PvP
- Clean RedMMO vs Titan session parity

Protected items

- Running animation / locomotion single-node guard on trooper ABP
- R92 package tree identity; fused 27/6 authoring hashes
- Do not enable both SIK and Epic OnlineSubsystemSteam together

Fab / plugins for this track

See FAB_MARKETPLACE_INVENTORY.md.

Need	Plugin
Netcode PIE fundamentals	None from Fab
Steam friend sessions	SteamIntegrationKit (project plugin; ~297 MB — may need copy from owner machine)
Planetary maps	PPG / PlanetGen — optional later

RedMMOTitan — Developer Handoff (2026-08-07)

Status: Good enough for developer takeover. Many problems remain. Planetary plugins are optional for first fundamentals work.

Ukrainian pack: START_HERE_UK.md · DEVELOPER_HANDOFF_UK.md · RedMMOTitan_Developer_Handoff_UK.pdf

Primary audience: programmer — networking, replication, multiplayer, PvP, gameplay. Start with NETWORKING_PVP.md / NETWORKING_PVP_UK.md.

Collaborator: sanyarud@gmail.com → GitHub sanyarud (write/push, pending accept on both private repos).

Artifact	Location
START_HERE	START_HERE.md · START_HERE_UK.md
PDF EN / UK	RedMMOTitan_Developer_Handoff.pdf · RedMMOTitan_Developer_Handoff_UK.pdf
Networking / PvP	NETWORKING_PVP.md · NETWORKING_PVP_UK.md
Fab inventory	FAB_MARKETPLACE_INVENTORY.md · FAB_MARKETPLACE_INVENTORY_UK.md
PPG-free start	PPG_PLANETGEN_FREE_START.md · PPG_PLANETGEN_FREE_START_UK.md
GitHub access	GITHUB_ACCESS.md
Diagnostics copy	D:\RedMMOTitanWindowsData\Diagnostics\RedMMO_DeveloperHandoff_20260807\
Canonical knowledge	D:\RedMMOTitan\ProjectKnowledge\ (INDEX.yaml → current_state.yaml → invariants.yaml)

1. Paths

Role	Path
Primary repo (owner, dirty worktree)	D:\RedMMOTitan — Titan.uproject
Fundamentals handoff project	TitanFundamentals.uproject (PlanetGen/PPG not required)
Clean planetary RedMMO	D:\RedMMOTitanWindowsData\Projects\RedMMO\RedMMO.uproject
Engine	D:\UE_5.8 (UE 5.8 exact)
R92 playable package	D:\RedMMOTitanWindowsData\Builds\RedMMO_R92_Playable_20260806T1740Z\Windows\RedMMO.exe
Tree SHA-256 (R92)	11F6F6AE332D56EE6BED0ADE800BFD72EC878990F2B262884FD162DD7445B409
Owner origin (private)	https://github.com/addgamestudios-ops/RedMMOTitan.git

Never use C:\Users\...\Documents\Red MM0 windows.

2. What works (honest)

Build / dependency (evidence: R92_DependencyUnblock pack):

- TitanEditor Win64 Development succeeds with FocalRig/WorldGen disabled and PlanetGen fork APIs shimmed (RedPlanetGenCompat).
- FocalRig: stock ControlRig AimItem substitute when Fab payload absent.
- WorldGen: disabled with documented rollback.

Live RedMMO editor PIE fixes (recent session — treat as working on that build, still need human regression):

- Mouse wheel → thrust
- Atmosphere exit
- Weapons / plumes / jetpack

- Terrain blend

Accepted packaged baseline:

- R92 Win64 Development package (50 files) — launchable without editor; PlayerStatus cook closure; packaged D3D12 smoke historically clean.

Still open / many problems remain:

- Human feel / full physical playtest on R92
- Authored HUD fuel pixels, audio feel, biome art polish
- Fused-continent consumer binding on clean RedMMO
- M04 traversal/perf, M05 real 1S/2C, Steam two-account, warp, final art
- Running anim must stay protected while improving feel

This handoff is good enough to take over, not “feature complete.”

3. Known bugs / open defects (pointers)

Read ProjectKnowledge/defects/ and queue Build/Automation/redmmotitan_module_queue.json. Snapshot from R92 takeover:

- DEF-0001, DEF-0002 open
- DEF-0003, DEF-0005, DEF-0007 await runtime acceptance
- DEF-0004 open (jetpack HUD boundary history)
- DEF-0006 awaits build/runtime
- DEF-0008 resolved
- UAIP / Copilot / Blender MCP: marketplace/human blockers — skip without stalling

4. FocalRig / WorldGen / PlanetGen disposition

Plugin	Disposition	Rollback
FocalRig	Disabled + ControlRig substitute	Restore Fab FocalRig.uplugin → Enabled true → rebuild
WorldGen	Disabled	Restore Wor ldGen . uplugin → Enabled true
PlanetGen fork pin	Absent; shims active	Restore Plugins/PlanetGenPinned_* or port APIs; shims auto-forward
PlanetGen Marketplace 1.7	Present on owner engine; disabled in TitanFundamentals	Enable only when doing Titan planetary maps
PPG	Clean RedMMO only; not required for fundamentals	Install Marketplace PPG for home world

Details: R92_DEPENDENCY_DISPOSITION.md and PPG_PLANETGEN_FREE_START.md.

5. Build / launch / PIE

Fundamentals (recommended first day)

```
# Open fundamentals project (no PPG/PlanetGen)
start D:\path\to\RedMMOTitan-DeveloperHandoff\TitanFundamentals.uproject
# Build TitanEditor Win64 Development, PIE on ThirdPersonMap
```

Full Titan (PlanetGen)

```
start D:\RedMMOTitan\Titan.uproject
# Map: /Game/RedMMO/Maps/RedPlanetGen
```

Clean planetary RedMMO (PPG)

```
start D:\RedMMOTitanWindowsData\Projects\RedMMO\RedMMO.uproject
# Map: /Game/RedMMO/Maps/RedMMO_PPG_HomeWorld - seed 1337
```

Packaged smoke (no editor)

```
& 'D:\RedMMOTitanWindowsData\Builds\RedMMO_R92_Playable_20260806T1740Z\Windows\RedMMO.exe'
```

Single-editor rule: one UnrealEditor at a time. Prefer guarded MCP start:

```
powershell -NoProfile -ExecutionPolicy Bypass -File D:\RedMMOTitan\Build\Automation\Start-UnrealAIEditorGuarded.ps1
```

6. MCP ports

Server	Port	Notes
Epic ModelContextProtocol	http://127.0.0.1:8000/mcp	Needs editor + -ModelContextProtocolStartServer
Nwiro	http://127.0.0.1:5353/mcp	Needs Nwiro plugins
UAIP	http://127.0.0.1:8765/mcp	Blocked — payload missing

Cursor catalog IDs: user-epic_unreal_mcp, user-nwiro, user-uaip_titan_http.

Guide: ... \RedMMO_R92_AgentTakeover_20260806T1805Z\MCP_PLUGIN_ACCESS_GUIDE.md.

7. Protected items

- 1 Running animation / locomotion graph on ABP_RedTrooperFemale — do not “fix” run by replacing the single-node / R86 locomotion guard without explicit owner authority. Prefer additive or parallel work on aim/jetpack poses.
- 2 Fused 27 tangent inputs + six fused faces — never silent bind/regenerate/promote.
- 3 Protected hashes (production map, 50 km checkpoint, fused prototype/heightfield, clean home, ProfileV1) — see takeover pack §6.
- 4 R92 package directory — immutable; new work gets a new named package.
- 5 Owner dirty git worktree — no hard reset / bulk clean.

8. Next fundamentals work

- 1 PPG/PlanetGen-free compile + PIE on TitanFundamentals / ThirdPersonMap.
- 2 Protect run anim; improve on-foot feel, aim, jetpack, fire separately.
- 3 Re-verify wheel-thrust, atmosphere exit, weapons/plumes on RedMMO when plugins available.
- 4 Only then: PPG home art, shore bands, fused consumer, MP/Steam.

9. GitHub

See GITHUB_ACCESS.md.

- Existing org/user repo addgamestudios-ops/RedMMOTitan is private.
- Standalone handoff repo is published separately for the developer.
- Inviting a collaborator to a private repo requires the developer's GitHub username (or email for invite). Agents cannot invent that handle.

10. Fab / Marketplace

Full checklist: FAB_MARKETPLACE_INVENTORY.md.

Day-1 rule: developer must not need PPG or PlanetGen installed to start building.

PPG / PlanetGen—free start (required for developer onboarding)

Goal: The developer must be able to build and PIE fundamentals without installing PPG or PlanetGen.

Verdict (2026-08-07)

Path	Needs PPG?	Needs PlanetGen?	Use for
TitanFundamentals.uproject + ThirdPersonMap	No	No	Immediate compile, controls, weapons, HUD, C++ fundamentals
Titan.uproject + RedPlanetGen	No (PPG not in Titan)	Yes (enabled; home map is PlanetGen)	Full Titan planetary prototype
Clean RedMMO.uproject + RedMMO_PPG_HomeWorld	Yes	Reserved / separate	Accepted R92 planetary baseline

Feasible in this handoff: Yes — use TitanFundamentals.uproject.

PlanetGen and PPG are optional for the fundamentals baseline. Full planet maps remain plugin-dependent and are documented below.

How to start without PPG / PlanetGen

- 1 Open TitanFundamentals.uproject (not RedMMO.uproject, not full planetary maps).
- 2 Confirm plugins:
 - PlanetGen → Enabled: false
 - WorldGen → Enabled: false
 - FocalRig → Enabled: false (stock ControlRig substitute already in source)
 - No PPG entry in this uproject
- 1 Editor / game default map: /Game/ThirdPerson/Maps/ThirdPersonMap
- 2 Build TitanEditor Win64 Development → PIE.

Shims already in project source

Shim	Path	Behavior when plugin/fork missing
PlanetGen query compat	Source/RedMMO/RedPlanetGenCompat.h/.cpp	Forwards to fork APIs if headers exist; else chunk/radial substitutes so TitanEditor links
FocalRig optional	Source/RedMMO/RedMMO.Build.cs + RedMMOEditorTools	REDMMO_WITH_MARKETPLACE_FOCALRIG=0 → stock ControlRig AimItem substitute
WorldGen	Titan.uproject / Fundamentals	Disabled; no hard module dependency

What still needs plugins (optional later)

Work	Required
Clean home world RedMMO_PPG_HomeWorld, ProfileV1 biomes, native PPG water/chunks	PPG marketplace plugin
Titan RedPlanetGen / 50 km PlanetGen maps, ACLMPlanet streaming	PlanetGen (engine Marketplace 1.7 and/or project pin)
Exact fork terrain stamp / MacroHeightfield APIs	Plugins/PlanetGenPinned_* (absent on live tree; shims cover link)

Fallback maps

Map	Plugin need	Notes
-----	-------------	-------

/Game/ThirdPerson/Maps/ThirdPersonMap	None	Primary PPG/PlanetGen-free starter (included in handoff Content)
/Game/RedMMO/Maps/Sandbox_DesertDemoSparkle_T01	Usually none for open	Local Titan sandbox; large; may reference dressed content
/Game/RedMMO/Maps/RedPlanetGen*	PlanetGen	Do not use for first-day fundamentals
/Game/RedMMO/Maps/RedMMO_PPG_HomeWorld	PPG	Clean D-drive project only

If ThirdPerson content is missing after clone, create File → New Level → Empty Open World / Basic and set it as Editor Startup Map under Project Settings. That empty level is enough to compile and test C++ pawn/GameMode wiring.

Restoring planetary plugins later

See FAB_MARKETPLACE_INVENTORY.md and

D:\RedMMOTitanWindowsData\Diagnostics\R92_DependencyUnblock_20260807_000731\R92_DEPENDENCY_DISPOSITION.md

.

- 1 Install official Fab payloads so *.uplugin exists (FocalRig/WorldGen currently Intermediate-only on owner machine).
- 2 Re-enable plugin flags in Titan.uproject.
- 3 Rebuild; FocalRig auto-links when descriptor returns.
- 4 For clean RedMMO planetary baseline, keep using D:\RedMMOTitanWindowsData\Projects\RedMMO\RedMMO.uproject with PPG enabled.

Fab / Marketplace inventory checklist

Purpose: What came from Fab/Marketplace, whether it is required for fundamentals, and what happens if missing.

Sources: Titan.uproject, RedMMO.uproject, Plugins/, ProjectKnowledge, CODEX_HANDOVER.md, R92 dependency disposition, MCP guide.

Legend: Req = required for fundamentals (PPG-free) · Planet = required for planetary maps · Opt = optional accelerator · Off = disabled / substituted now

Plugins / engine marketplace

Name	Purpose	Fundamentals	If missing	Fab / notes
PlanetGen	Spherical planet (AC LMP planet), Titan RedPlanetGen maps	Off for fundamentals; Planet for Titan maps	TitanFundamentals disables it. Live Titan enables Marketplace 1.7 at D:\UE_5.8\Engine\Plugins\Marketplace\PlanetGe5e8f23ed72d1V5\ . Pinned fork Plugins\PlanetGenPinned_1_4_0_RedMMO absent; RedPlanetGenCompat shims link without fork APIs	Fab/Epic Marketplace "PlanetGen" / CLM — install via Epic Launcher into UE 5.8
PPG (Procedural Planet Generator)	Clean RedMMO home world, ProfileV1, chunks/water/foilage	Off for fundamentals; Planet for clean RedMMO	Clean RedMMO.uproject will not generate home world. Fundamentals uproject has no PPG entry	Engine path seen: ... \Marketplace\Procedur890d9e860517V2\PPG.uplugin
FocalRig	Rifle aim chain / Aim Weapon ControlRig units	Off — substituted	Plugin Enabled: false. Marketplace folder Intermediate-only (no FocalRig.uplugin). Stock ControlRig AimItem substitute preserves FocalAimTarget / FocalAimWeight	Restore official Fab payload → enable in uproject → rebuild (Build.cs auto-links)
WorldGen	World-gen marketplace companion	Off	Enabled: false; Intermediate-only; documented rollback	Same as FocalRig — restore WorldGen.uplugin then enable
Nwirol + NwirolIntegrationKit	Editor MCP / environment tooling (: 5353)	Opt	Editor tooling only; not needed for packaged play or fundamentals PIE	MarketplaceURL product 22e3bc90-ed91-4154-8bea-21c33e6a51b7 (Epic Fab deep link in Titan.uproject)
UnrealAllIntegrationPlatform (UAIP)	UAIP MCP (: 8765)	Opt / blocked	Enabled: false; payload Intermediate-only; skip Fab stall	Fab: https://www.fab.com/listings/0eedf909-00ac-4d95-b109-8fda51800fff
UAIPMCPBridge	Stdio bridge that may auto-launch editor	Opt / blocked until UAIP	Present as bridge-only under Marketplace	Same UAIP listing
ModelContextProtocol + ToolsetRegistry / EditorToolset	Official Epic Unreal MCP (: 8000)	Opt	Engine Experimental plugin; start guarded editor to listen	Engine: D:\UE_5.8\Engine\Plugins\Experimental\ModelContextProtocol\
VibeMMOUKit	VibeMMO player HUD kit (project Plugins/)	Opt (HUD)	HUD may fall back / miss widgets; gameplay C++ can still run	Project plugin (from Mac/Vibe handoff); keep with project
RedHUD	Project HUD plugin	Opt / project	Status texture / fuel pixels incomplete without content	Project-owned under Plugins/RedHUD

SteamIntegrationKit	Steam OSS / sessions (SIK v1.9, App 480)	Opt (MP later)	Multiplayer lobby/Steam transport unavailable; PIE listen-server still possible with NULL OSS	Project Plugins/SteamIntegrationKit (~297 MB) — transfer deliberately
VibeUE	Vibe tooling	Off on Win64	PlatformDenylist: Win64	Marketplace VibeUE581860d1833205V1
BpGeneratorUltimate (Ultimate Engine Copilot)	BP generator MCP	Off on Win64	Denied Win64; no MCP surface	Skip
OnlineSubsystemSteam / SteamSockets (Epic)	Legacy Steam	Off	Replaced by SIK	Keep disabled

Content packs (Fab / Marketplace / purchased) — high level

Pack / area	Purpose	Fundamentals	If missing
Action Trooper / Tall Female (Content/Action_Trooper, characters)	Playable trooper mesh/anim	Needed for full infantry feel; ThirdPerson pawn works without	Use ThirdPerson template character first
StarSparrow	Modular ship	Planet/ship work	Skip until planetary session
Projectiles Vol.1 / Muzzles / BeamsPack	Weapon VFX, grapple cable	Nice-to-have for weapons polish	Bolt tracer C++ still compiles
Jet_Packs_Sci-Fi	Jetpack meshes/VFX	Jetpack presentation	Logic may exist without mesh
SoStylized / Stylized Desert Oasis / Stylized Rocks / TropicalAlienWorld / Alien_* packs	Biome art	Planet art only	Not needed for fundamentals map
Ultimate Stylized UI Crosshair	Arc sight	HUD polish	https://www.fab.com/listings/707a878d-0d11-431a-813c-a7aad10109f
SciFi_Skills_Icon	Ability icons for Vibe HUD	HUD	Cook directories note in CODEX_HANDOVER
SpaceColony / AsteroidSpaceport / Cloudz_Hi5 / Vefects / etc.	Environment dressing	No	Exclude from fundamentals

Size note: Full Content/ on owner machine is tens of GB and is not all in git. Handoff repo ships ThirdPerson + git-whitelisted RedMMO crumbs + docs/source. Remaining packs travel via drive / separate sync.

Quick “install only if you need...”

You want...	Install
Day-1 C++ / PIE / controls	Nothing from Fab — use TitanFundamentals.uproject
Clean R92 planetary home	PPG + clean project Content under WindowsData\Projects\RedMMO
Titan RedPlanetGen planet	PlanetGen UE 5.8 Marketplace
Official rifle aim rig	FocalRig full Fab payload (not Intermediate leftovers)
Epic/Nwiro MCP agents	Guarded editor + Nwiro (+ UAIP only after Fab reinstall)
Steam friend sessions	SteamIntegrationKit + Steam client + App 480

Owner-machine disposition snapshot (2026-08-07)

From R92_DEPENDENCY_DISPOSITION.md:

- FocalRig / WorldGen: not recoverable as official payloads → substituted / disabled
- PlanetGen pinned fork: absent; Marketplace 1.7 present; APIs shimmed
- TitanEditor Win64 Development: Succeeded after dependency unblock
- Sibling sessions may hold MCP on ports 8000 / 1985 when editor is up

GitHub access — standalone handoff + invite rules

Intended collaborator

Field	Value
Contact email	sanyarud@gmail.com
Resolved GitHub username	sanyarud (https://github.com/sanyarud , name: Oleksandr)
Permission	write / push (not read-only)
API note	Collaborator PUT requires username; email local-part matched sanyarud

Invitation status (sent 2026-08-06 UTC)

Repo	Invite ID	Permissions	Status	Accept URL
addgamestudios-ops/RedMMOTitan-DeveloperHandoff	328309879	write	pending accept (expired: false)	https://github.com/addgamestudios-ops/RedMMOTitan-DeveloperHandoff/invitations
addgamestudios-ops/RedMMOTitan	328309881	write	pending accept (expired: false)	https://github.com/addgamestudios-ops/RedMMOTitan/invitations

Developer must accept while logged into GitHub as sanyarud (GitHub notifies that account's email).

Answer (invite vs open)

Repo	Visibility	Must invite by username?
addgamestudios-ops/RedMMOTitan	Private	YES — done for sanyarud (from sanyarud@gmail.com)
addgamestudios-ops/RedMMOTitan-DeveloperHandoff	Private	YES — done for sanyarud

Re-invite if needed:

```
gh api -X PUT repos/addgamestudios-ops/RedMMOTitan-DeveloperHandoff/collaborators/sanyarud -f permission=push
gh api -X PUT repos/addgamestudios-ops/RedMMOTitan/collaborators/sanyarud -f permission=push
```

Clone URL (standalone handoff)

<https://github.com/addgamestudios-ops/RedMMOTitan-DeveloperHandoff.git>

```
git clone https://github.com/addgamestudios-ops/RedMMOTitan-DeveloperHandoff.git
cd RedMMOTitan-DeveloperHandoff
# Read Docs/DeveloperHandoff/START_HERE.md
start TitanFundamentals.uproject
```

Web UI: <https://github.com/addgamestudios-ops/RedMMOTitan-DeveloperHandoff>

What is included vs excluded

Included in standalone handoff repo

- Source/ (RedMMO gameplay C++)

- Config/
- docs/ + Docs/DeveloperHandoff/ (MD + PDF)
- ProjectKnowledge/
- Build/Automation/ (scripts that exist)
- Selected Tools/ (no __pycache__)
- Titan.uproject, TitanFundamentals.uproject, targets
- Content/ThirdPerson/ (PPG-free starter map)
- Git-whitelisted Content/RedMM0/ crumbs (maps/materials/characters listed in .gitignore exceptions) when present
- Project plugins that are small enough / essential: prefer VibeMMOUIKit, RedHUD if size allows; otherwise document copy-from-owner
- README / CODEX_HANDOVER / WINDOWS_BUILD entry docs

Excluded (do not expect in clone)

- Binaries/, Intermediate/, DerivedDataCache/, Saved/
- Full Fab content packs (TropicalAlienWorld, SpaceColony, Action_Trooper full tree, etc. — tens of GB)
- Git LFS large media beyond tracked exceptions (clone may need separate content drive)
- Plugins/SteamIntegrationKit (~297 MB) — optional; copy from owner machine for Steam work
- .env, credentials, Steam secrets beyond public App ID 480 for Spacewar tests
- Owner-only diagnostics under D:\RedMM0TitanWindowsData\ (R92 exe, rollbacks) — documented by path, not uploaded

Existing private Titan remote

- <https://github.com/addgamestudios-ops/RedMM0Titan.git> — historical Mac/Windows handoff; worktree on owner PC is intentionally dirty and not fully mirrored by a blind push of all Content.

If gh create/push failed on a machine

```
gh auth login
gh repo create addgamestudios-ops/RedMM0Titan-DeveloperHandoff --private --source=. --remote=origin --push
# then invite:
gh api -X PUT repos/addgamestudios-ops/RedMM0Titan-DeveloperHandoff/collaborators/DEVELOPER_GITHUB_USERNAME -f permission=p
```